Pikachu Metrics 数据分析 metrics_Mjlab-Tracking-Flat-Pikachu-27dof-Walk__model_4999_20260910_165340_with_target

FILE
metrics_Mjlab-Tracking-Flat-Pikachu-27dof-Walk__model_4999_20260910_165340_with_target.csv

NETWORK
Mjlab-Tracking-Flat-Pikachu-27dof-Walk__model_4999

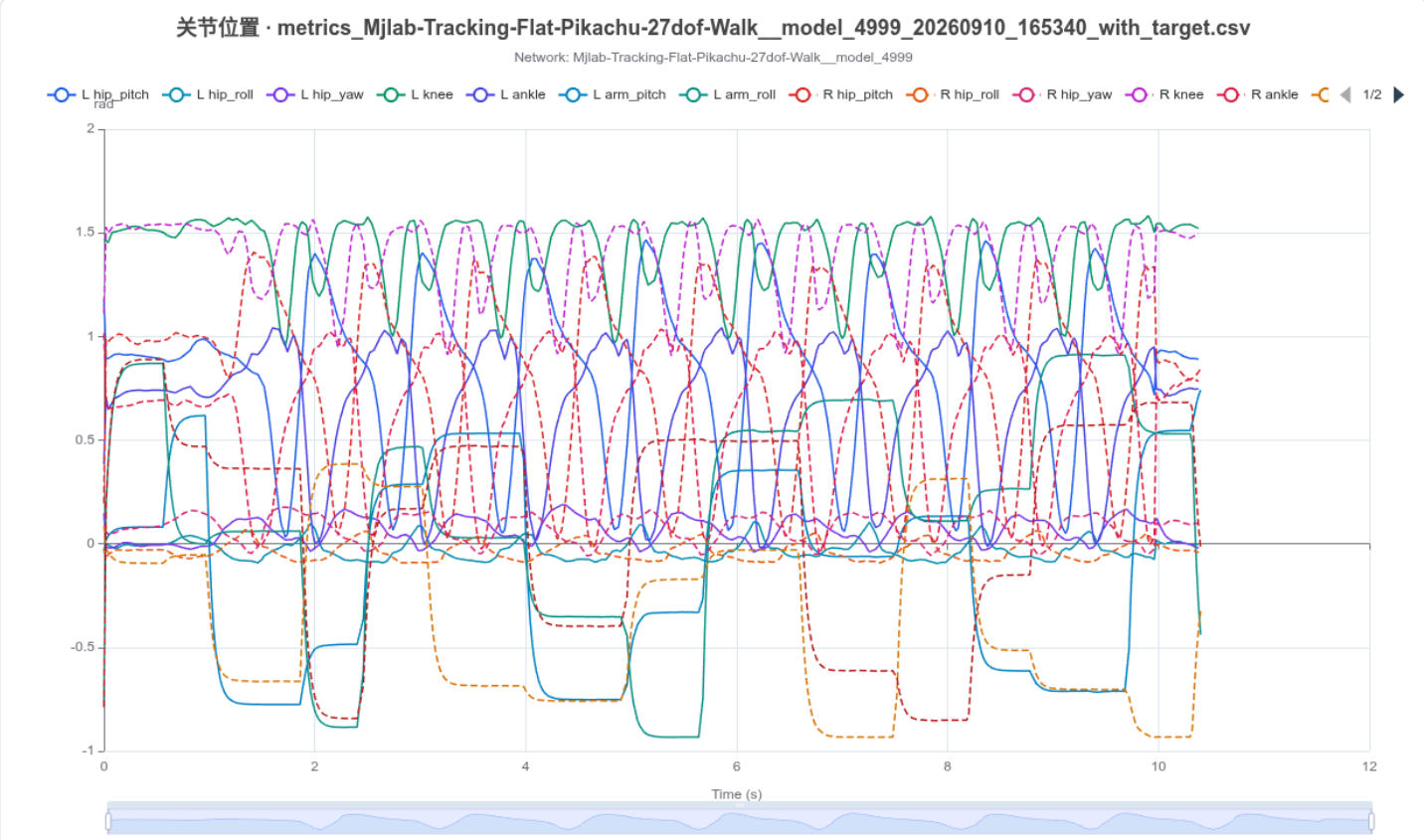
SAMPLES
454

DURATION
10.400 s

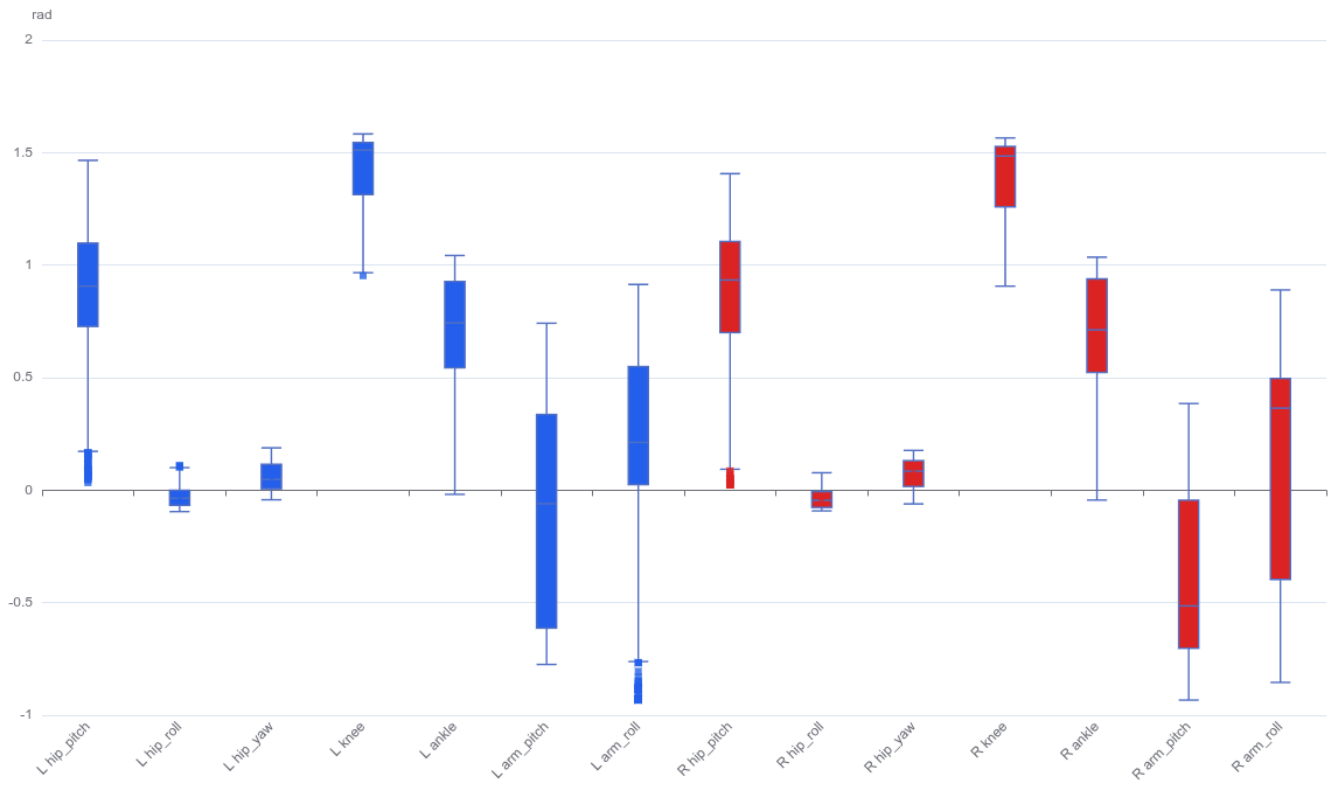
ENVIRONMENTS
1

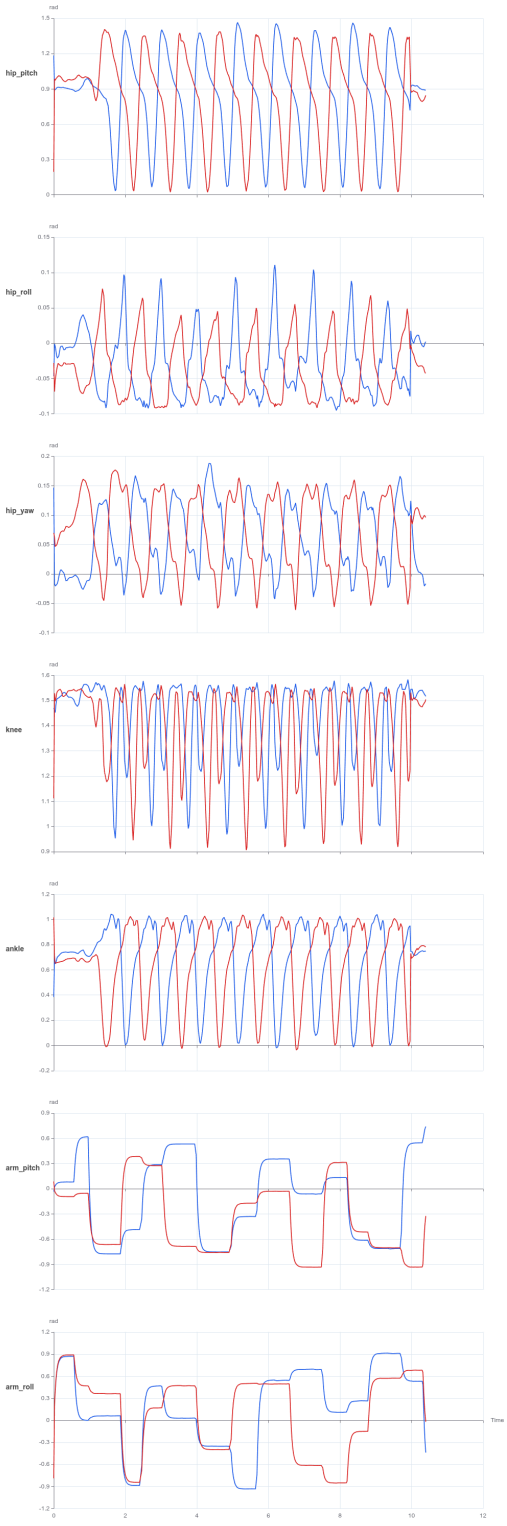
时间区间: 0.000 — 10.400 s · 仅含已勾选字段

关节位置

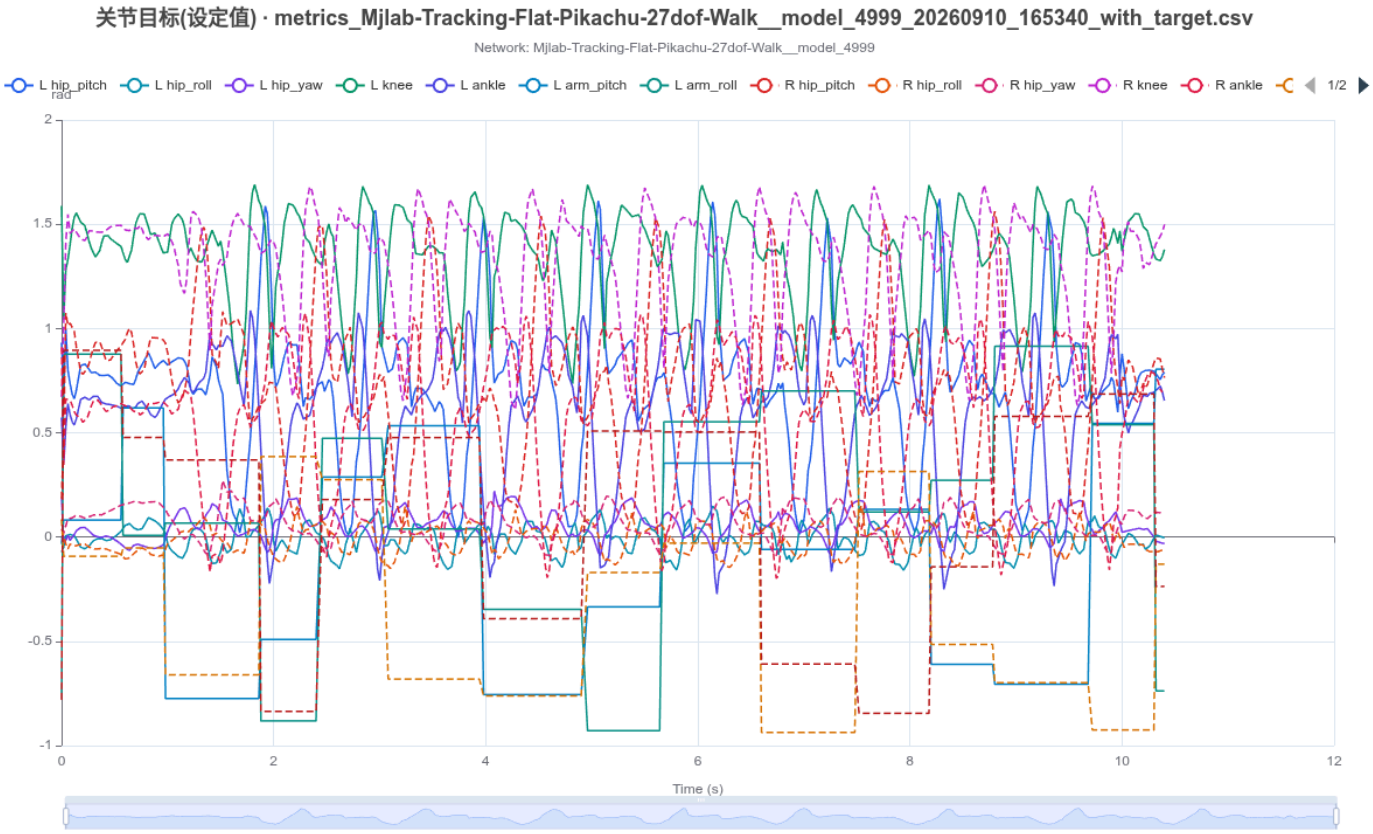


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_position_rad/left_hip_pitch_joint	30	10	454	0.0333248	1.46536	0.867559	0.942114	1.46536	1.39737
joint_position_rad/left_hip_roll_joint	10	15	454	-0.0951254	0.110861	-0.0279001	0.0553648	0.110861	0.0895976
joint_position_rad/left_hip_yaw_joint	10	15	454	-0.0425950	0.187962	0.0605338	0.0850529	0.187962	0.149308
joint_position_rad/left_knee_joint	30	10	454	0.953010	1.58284	1.42534	1.43472	1.58284	1.56400
joint_position_rad/left_ankle_joint	10	15	454	-0.0189360	1.04290	0.679703	0.744163	1.04290	1.01278
joint_position_rad/left_arm_pitch_joint	10	15	454	-0.774713	0.741454	-0.113349	0.494418	0.774713	0.769521
joint_position_rad/left_arm_roll_joint	10	15	454	-0.933248	0.913959	0.192552	0.556043	0.933248	0.913737
joint_position_rad/right_hip_pitch_joint	30	10	454	0.0226461	1.40641	0.858070	0.937229	1.40641	1.35348
joint_position_rad/right_hip_roll_joint	10	15	454	-0.0920286	0.0771898	-0.0361553	0.0564722	0.0920286	0.0882672
joint_position_rad/right_hip_yaw_joint	10	15	454	-0.0612577	0.176773	0.0731377	0.0979475	0.176773	0.153177
joint_position_rad/right_knee_joint	30	10	454	0.906457	1.56486	1.38821	1.39960	1.56486	1.54422
joint_position_rad/right_ankle_joint	10	15	454	-0.0439892	1.03521	0.660744	0.730033	1.03521	1.00700
joint_position_rad/right_arm_pitch_joint	10	15	454	-0.932973	0.385026	-0.382125	0.569627	0.932973	0.931798
joint_position_rad/right_arm_roll_joint	10	15	454	-0.854297	0.889464	0.115058	0.543435	0.889464	0.850786

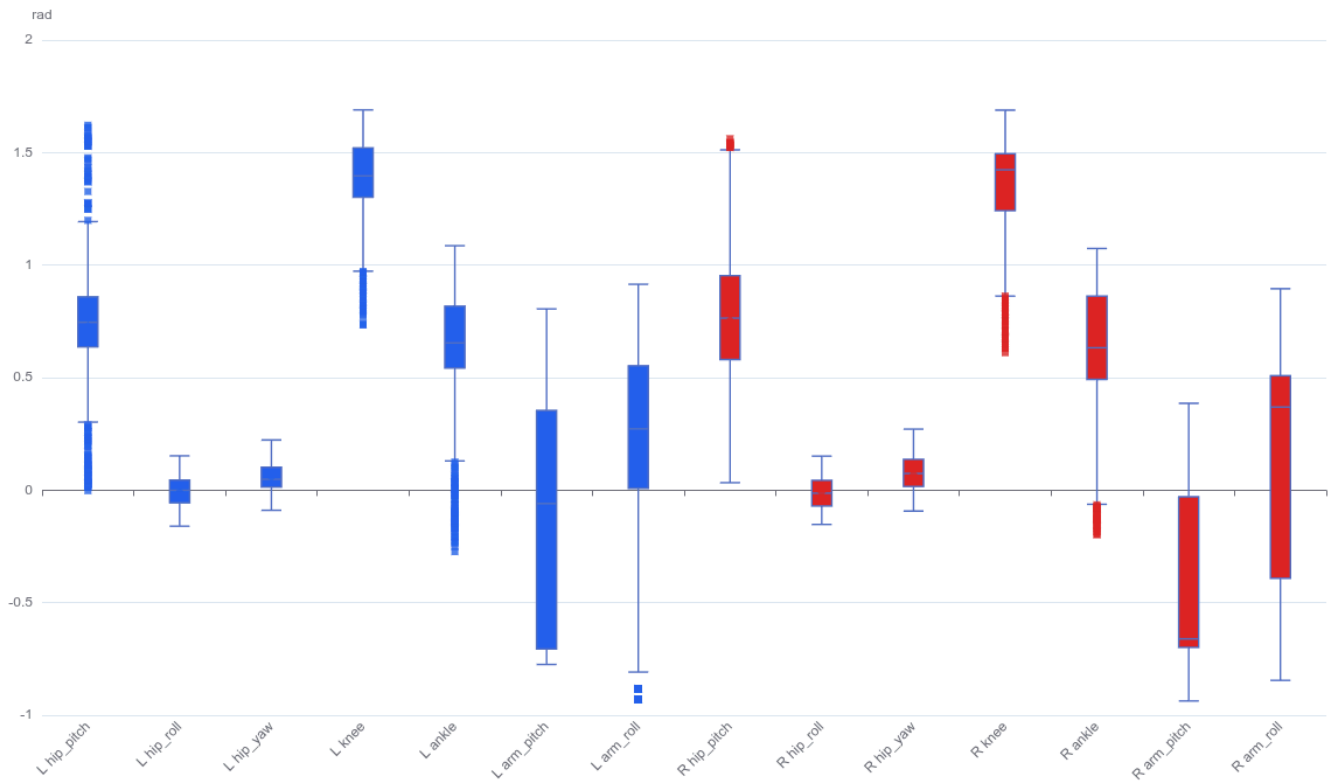

rad
2
1.5
1
0.5
0
-0.5
-1
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

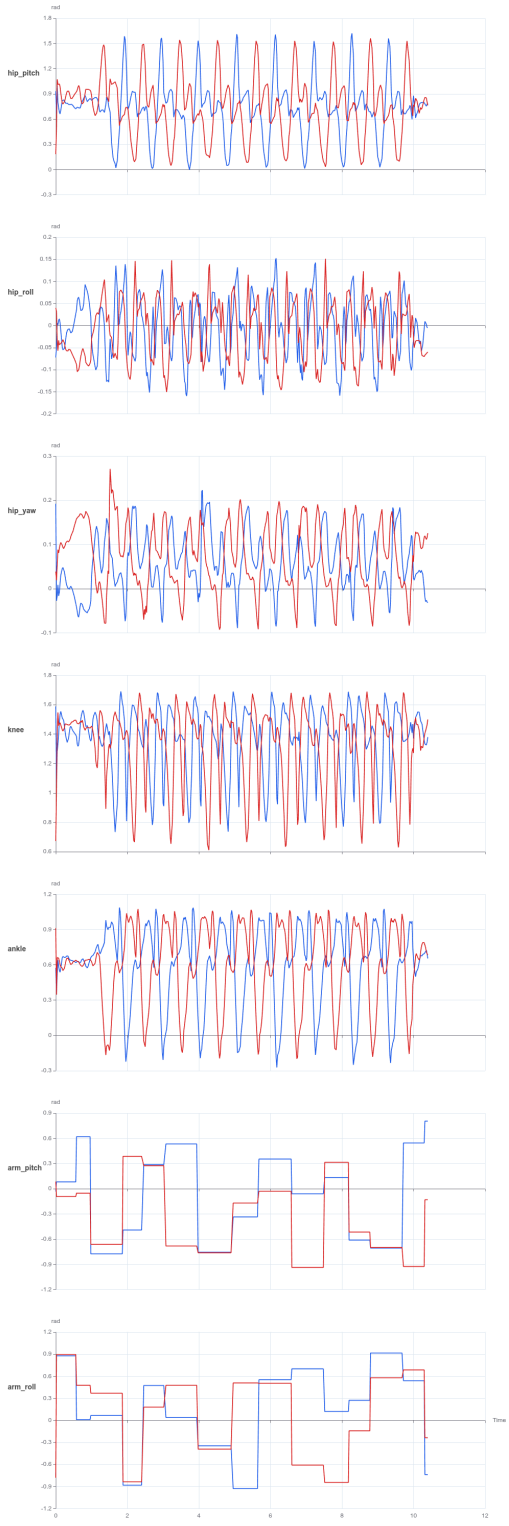

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
Time

关节目标(设定值)

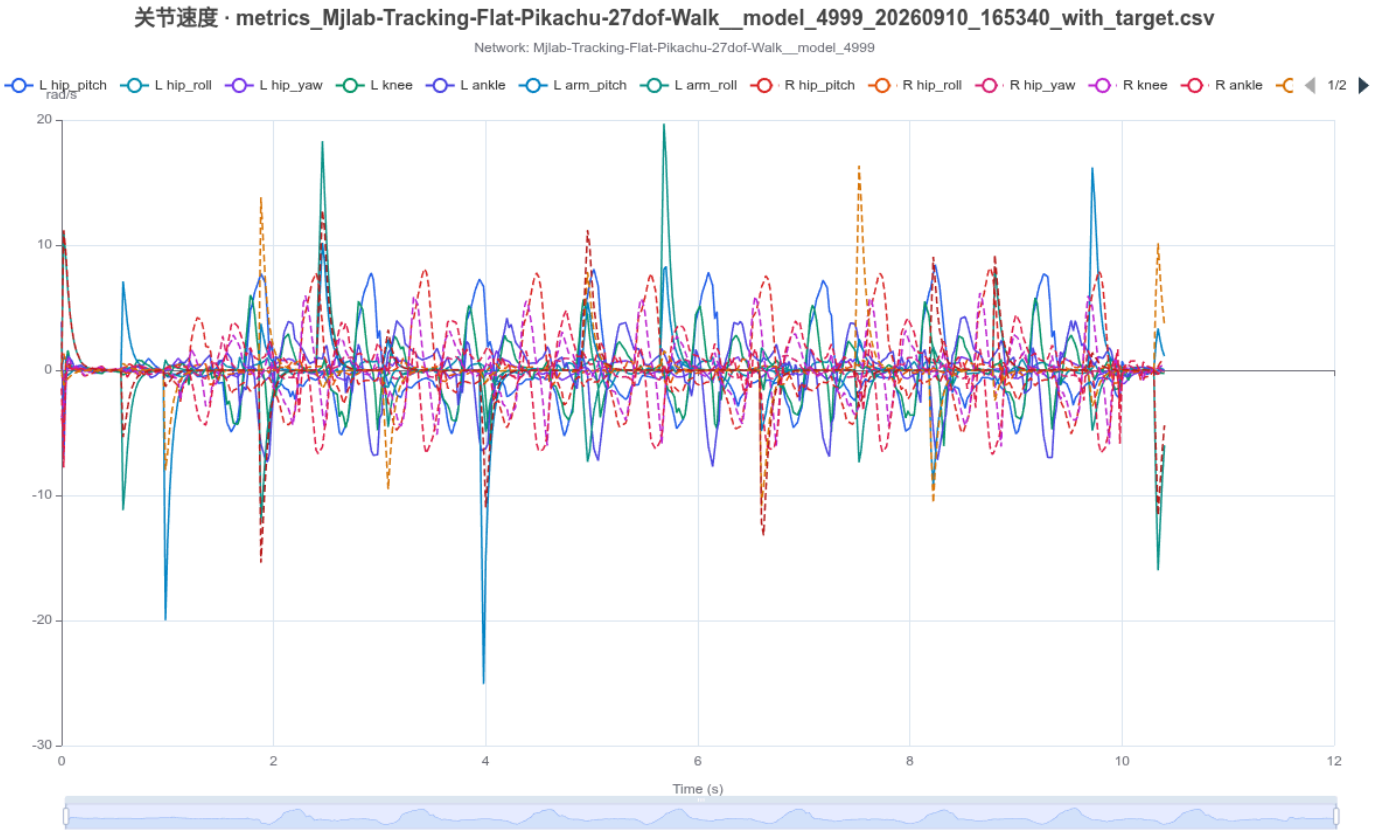


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_target_rad/left_hip_pitch_joint	30	10	454	-0.00237497	1.62159	0.731309	0.802302	1.62159	1.40197
joint_target_rad/left_hip_roll_joint	10	15	454	-0.160115	0.152356	-0.00467140	0.0677913	0.160115	0.128185
joint_target_rad/left_hip_yaw_joint	10	15	454	-0.0895209	0.222926	0.0572927	0.0869041	0.222926	0.175168
joint_target_rad/left_knee_joint	30	10	454	0.733665	1.68979	1.36025	1.37803	1.68979	1.62848
joint_target_rad/left_ankle_joint	10	15	454	-0.273185	1.08638	0.601084	0.685274	1.08638	1.00169
joint_target_rad/left_arm_pitch_joint	10	15	454	-0.774494	0.805069	-0.111314	0.520057	0.805069	0.774494
joint_target_rad/left_arm_roll_joint	10	15	454	-0.928408	0.915059	0.196735	0.583164	0.928408	0.928408
joint_target_rad/right_hip_pitch_joint	30	10	454	0.0334447	1.56188	0.757095	0.835287	1.56188	1.45144
joint_target_rad/right_hip_roll_joint	10	15	454	-0.152210	0.151394	-0.0118519	0.0693191	0.152210	0.122102
joint_target_rad/right_hip_yaw_joint	10	15	454	-0.0929882	0.271311	0.0732390	0.105009	0.271311	0.182875
joint_target_rad/right_knee_joint	30	10	454	0.610764	1.68852	1.32815	1.35325	1.68852	1.62115
joint_target_rad/right_ankle_joint	10	15	454	-0.198569	1.07387	0.595076	0.681954	1.07387	1.01465
joint_target_rad/right_arm_pitch_joint	10	15	454	-0.936729	0.385852	-0.383242	0.584666	0.936729	0.936729
joint_target_rad/right_arm_roll_joint	10	15	454	-0.845247	0.895102	0.118357	0.562801	0.895102	0.895102

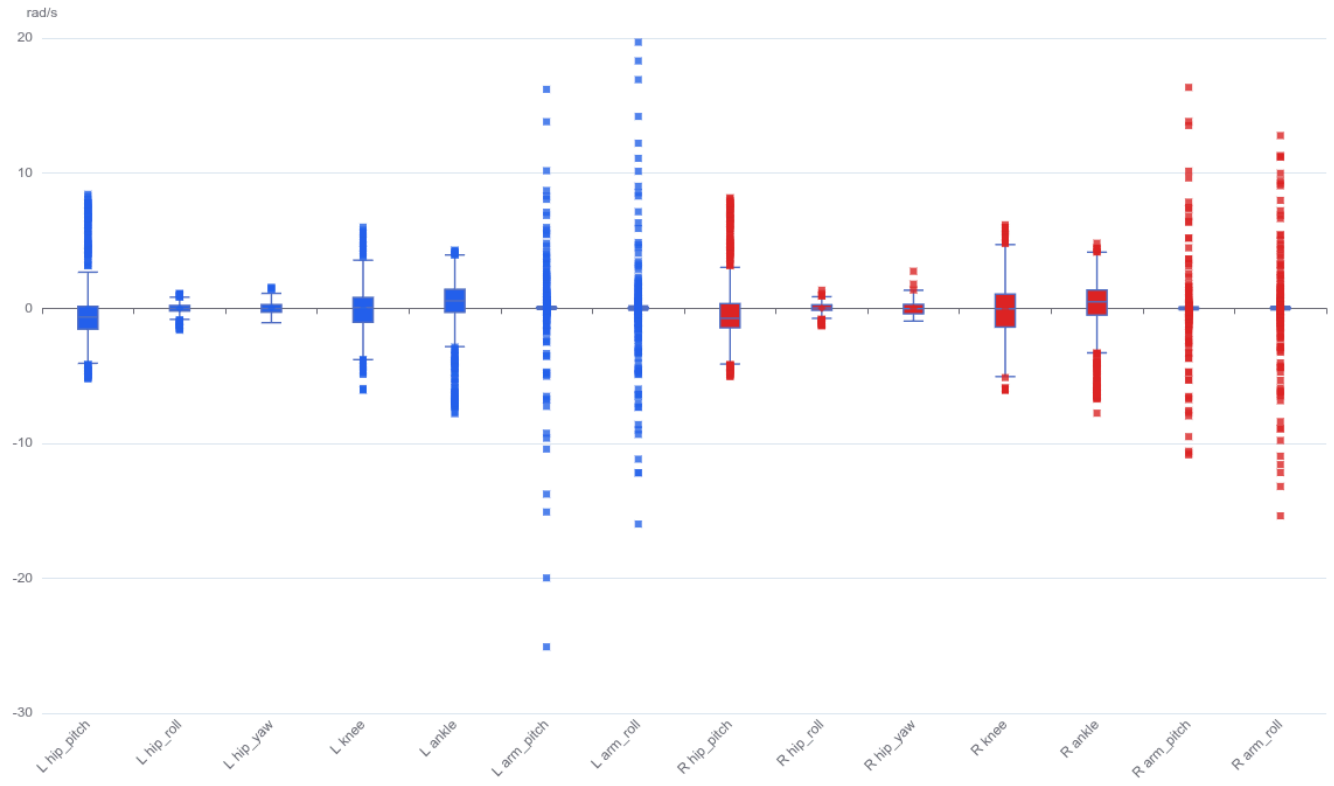

rad
2
1.5
1
0.5
0
-0.5
-1
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

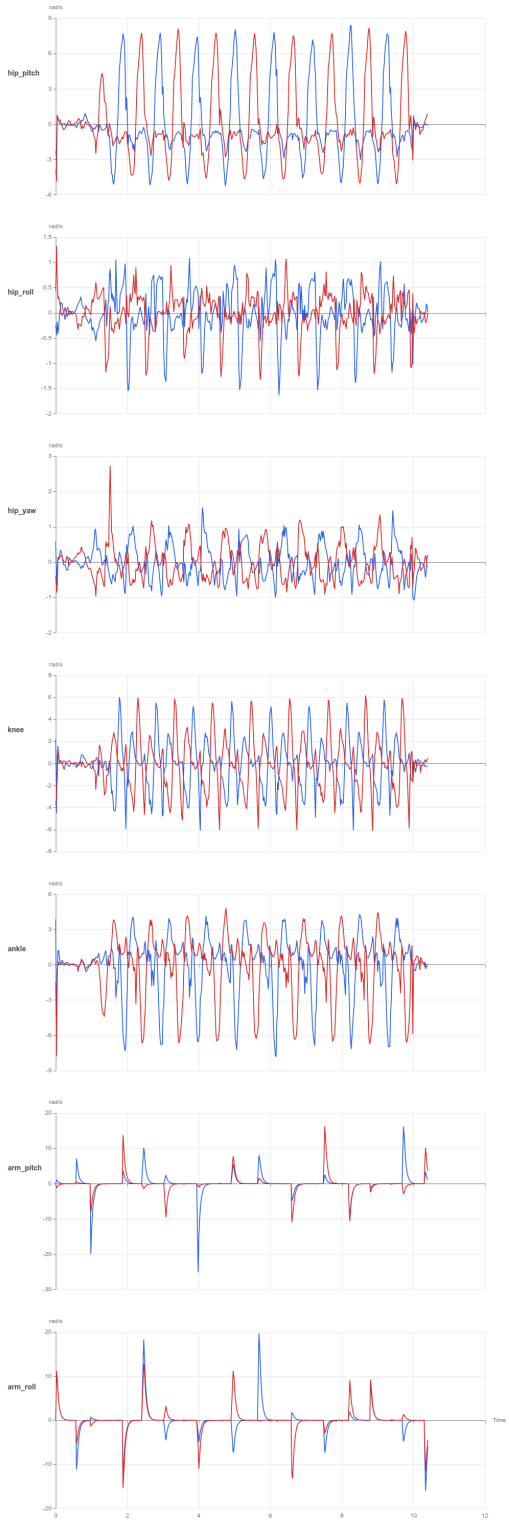

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
rad
Time

关节速度

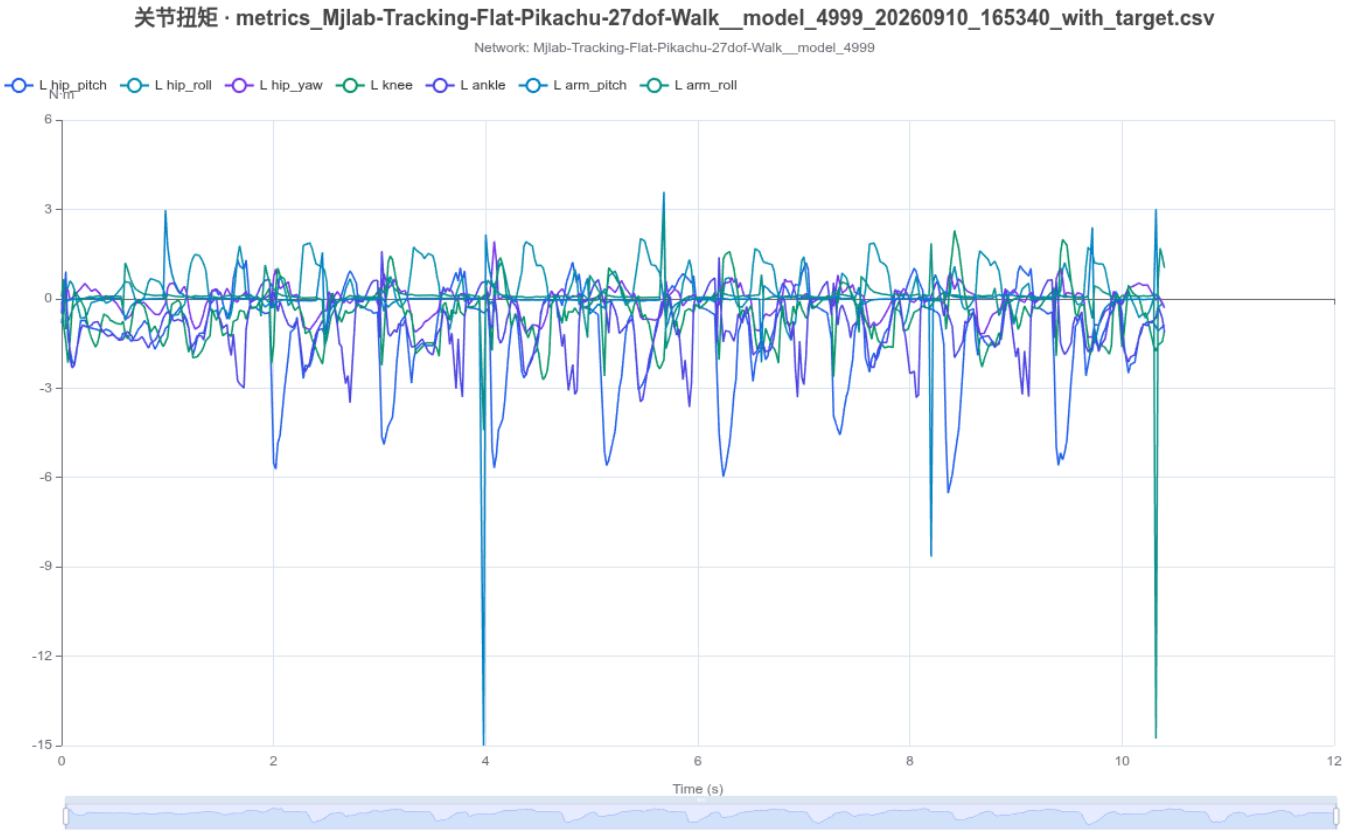


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_velocity_rad_s/left_hip_pitch_joint	30	10	454	-5.24973	8.42830	-0.0570740	2.96442	8.42830	6.75059
joint_velocity_rad_s/left_hip_roll_joint	10	15	454	-1.63370	1.09237	-0.0221783	0.473779	1.63370	1.09009
joint_velocity_rad_s/left_hip_yaw_joint	10	15	454	-1.07889	1.54800	0.00621719	0.467377	1.54800	0.927131
joint_velocity_rad_s/left_knee_joint	30	10	454	-6.06944	6.00003	-0.0290068	2.10330	6.06944	4.59894
joint_velocity_rad_s/left_ankle_joint	10	15	454	-7.81745	4.27719	0.0695659	2.47587	7.81745	6.28416
joint_velocity_rad_s/left_arm_pitch_joint	10	15	454	-25.0854	16.2020	0.0212558	2.72074	25.0854	5.04187
joint_velocity_rad_s/left_arm_roll_joint	10	15	454	-15.9896	19.6940	-0.0231053	2.83477	19.6940	6.38797
joint_velocity_rad_s/right_hip_pitch_joint	30	10	454	-5.08164	8.17127	-0.0379213	2.98542	8.17127	6.71393
joint_velocity_rad_s/right_hip_roll_joint	10	15	454	-1.32341	1.32828	0.00169721	0.392469	1.32828	0.932267
joint_velocity_rad_s/right_hip_yaw_joint	10	15	454	-0.963699	2.72942	-0.00585912	0.499250	2.72942	0.963699
joint_velocity_rad_s/right_knee_joint	30	10	454	-6.10305	6.17297	-0.0769882	2.30512	6.17297	5.05678
joint_velocity_rad_s/right_ankle_joint	10	15	454	-7.77816	4.80602	-0.0267814	2.51284	7.77816	6.08343
joint_velocity_rad_s/right_arm_pitch_joint	10	15	454	-10.8711	16.3471	-0.0290670	2.23590	16.3471	5.20247
joint_velocity_rad_s/right_arm_roll_joint	10	15	454	-15.3820	12.7789	-0.0334425	2.55819	15.3820	6.33407

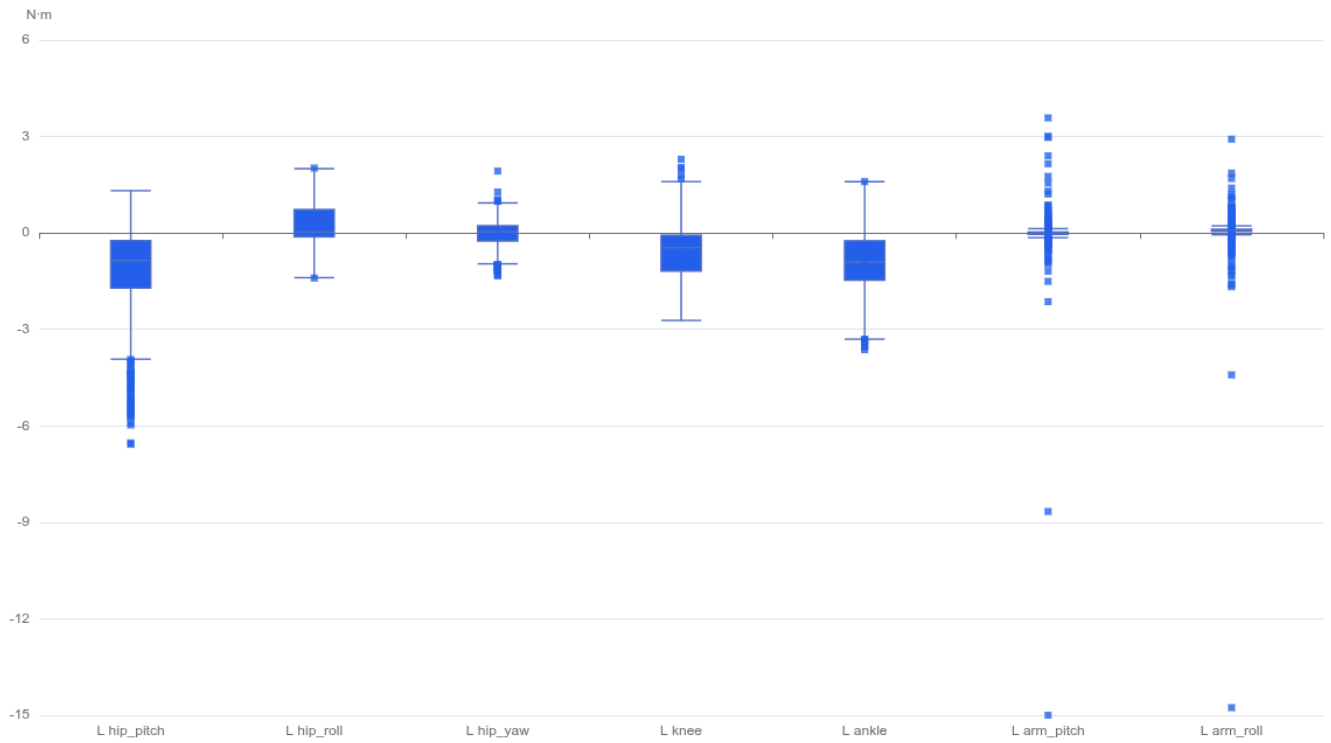

rad/s
20
10
0
-10
-20
-30
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

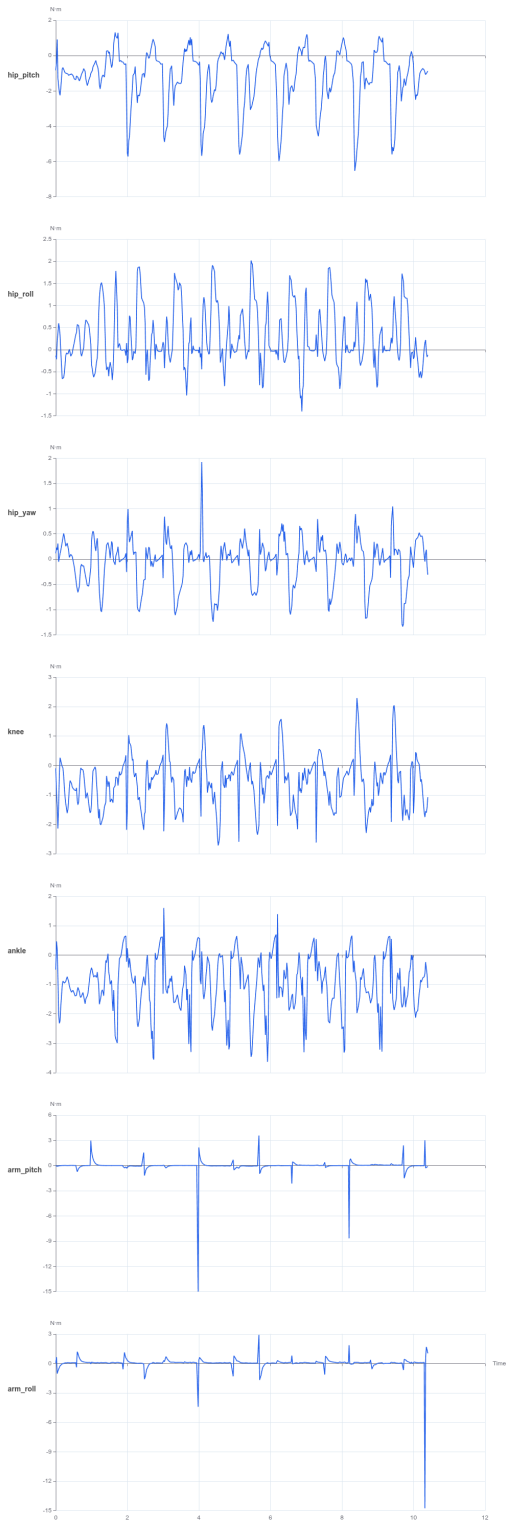

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
rad/s
Time

关节扭矩

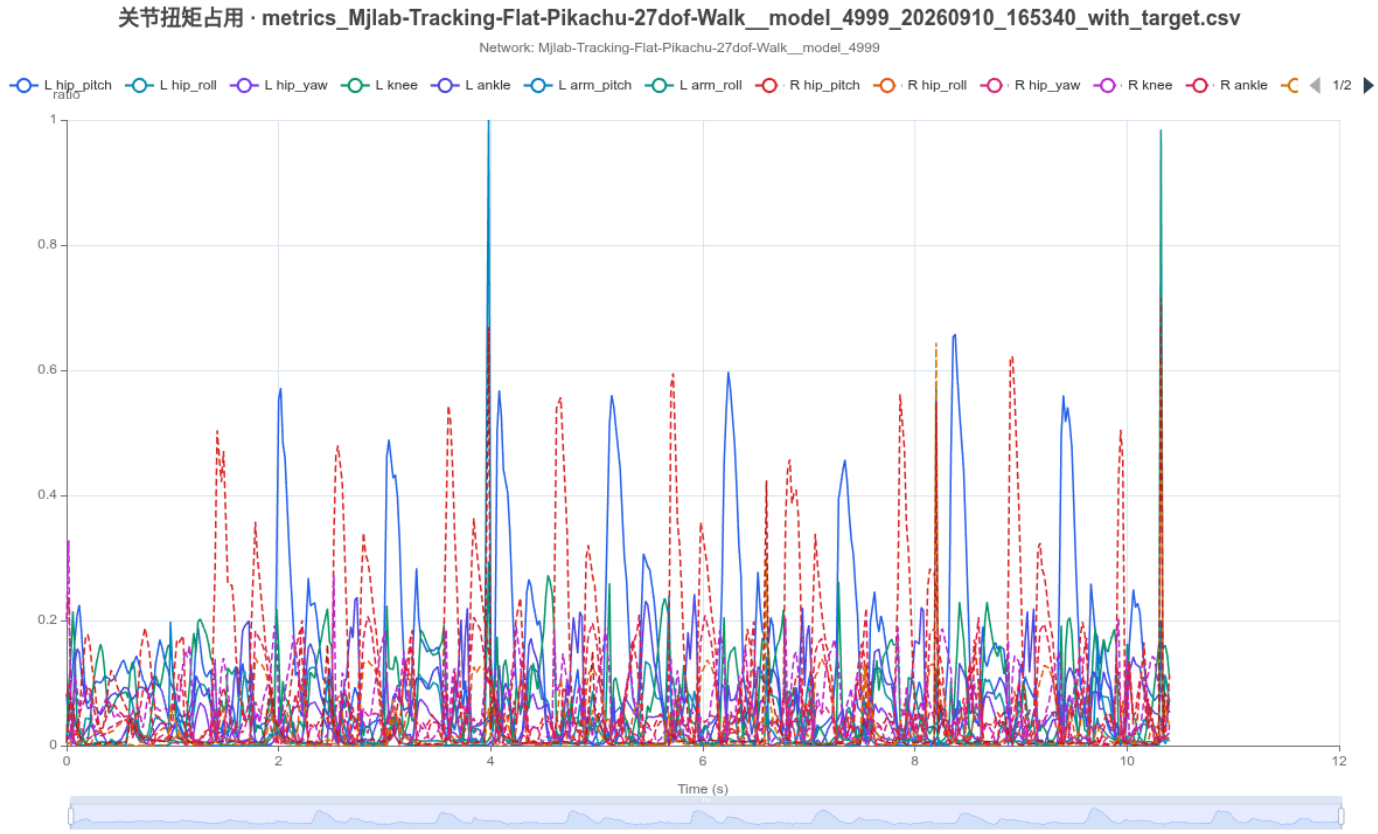


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_torque_nm/left_hip_pitch_joint	30	10	454	-6.57379	1.31399	-1.16600	1.96187	6.57379	4.81603
joint_torque_nm/left_hip_roll_joint	10	15	454	-1.40276	2.01885	0.283333	0.730668	2.01885	1.60799
joint_torque_nm/left_hip_yaw_joint	10	15	454	-1.33749	1.92523	-0.0408238	0.463940	1.92523	1.00438
joint_torque_nm/left_knee_joint	30	10	454	-2.72034	2.29272	-0.559902	1.01947	2.72034	1.98074
joint_torque_nm/left_ankle_joint	10	15	454	-3.63243	1.59673	-0.944699	1.32205	3.63243	2.71669
joint_torque_nm/left_arm_pitch_joint	10	15	454	-15.0000	3.57842	-0.0286773	0.904753	15.0000	0.641535
joint_torque_nm/left_arm_roll_joint	10	15	454	-14.7652	2.91734	0.0475343	0.803938	14.7652	0.775410

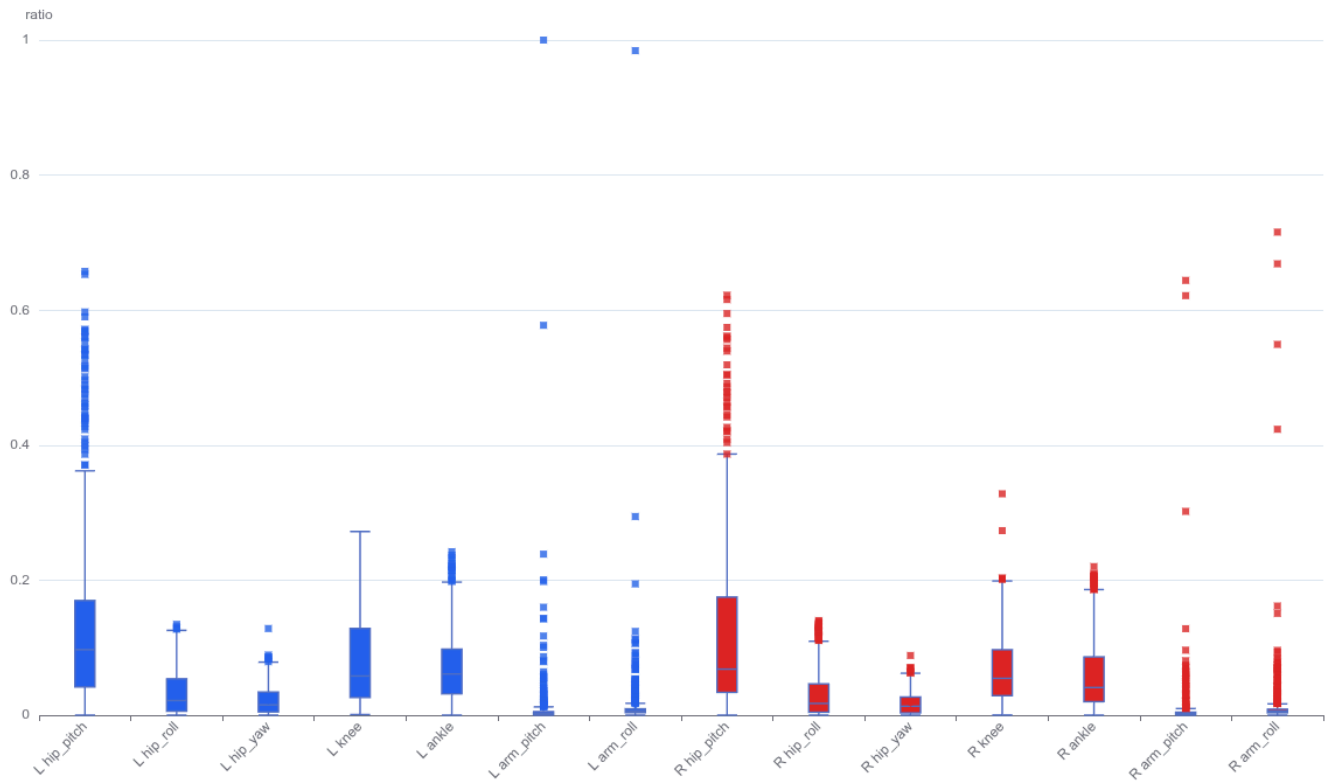

N·m
6
3
0
-3
-6
-9
-12
-15
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll

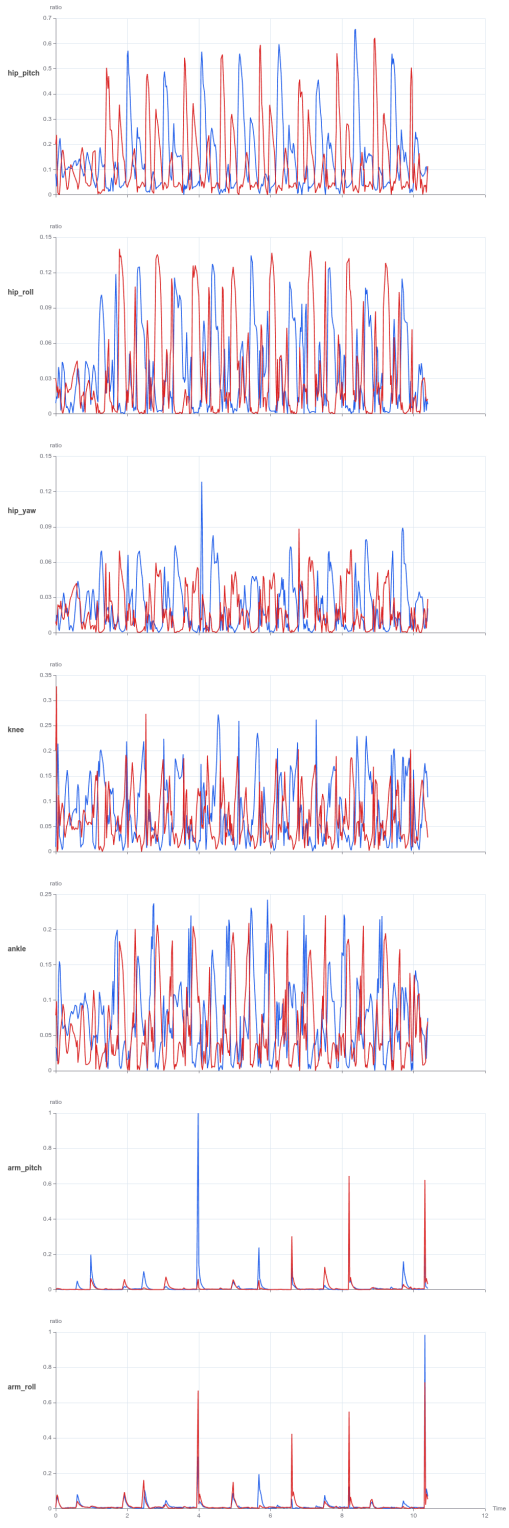

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
N·m
Time

关节扭矩占用

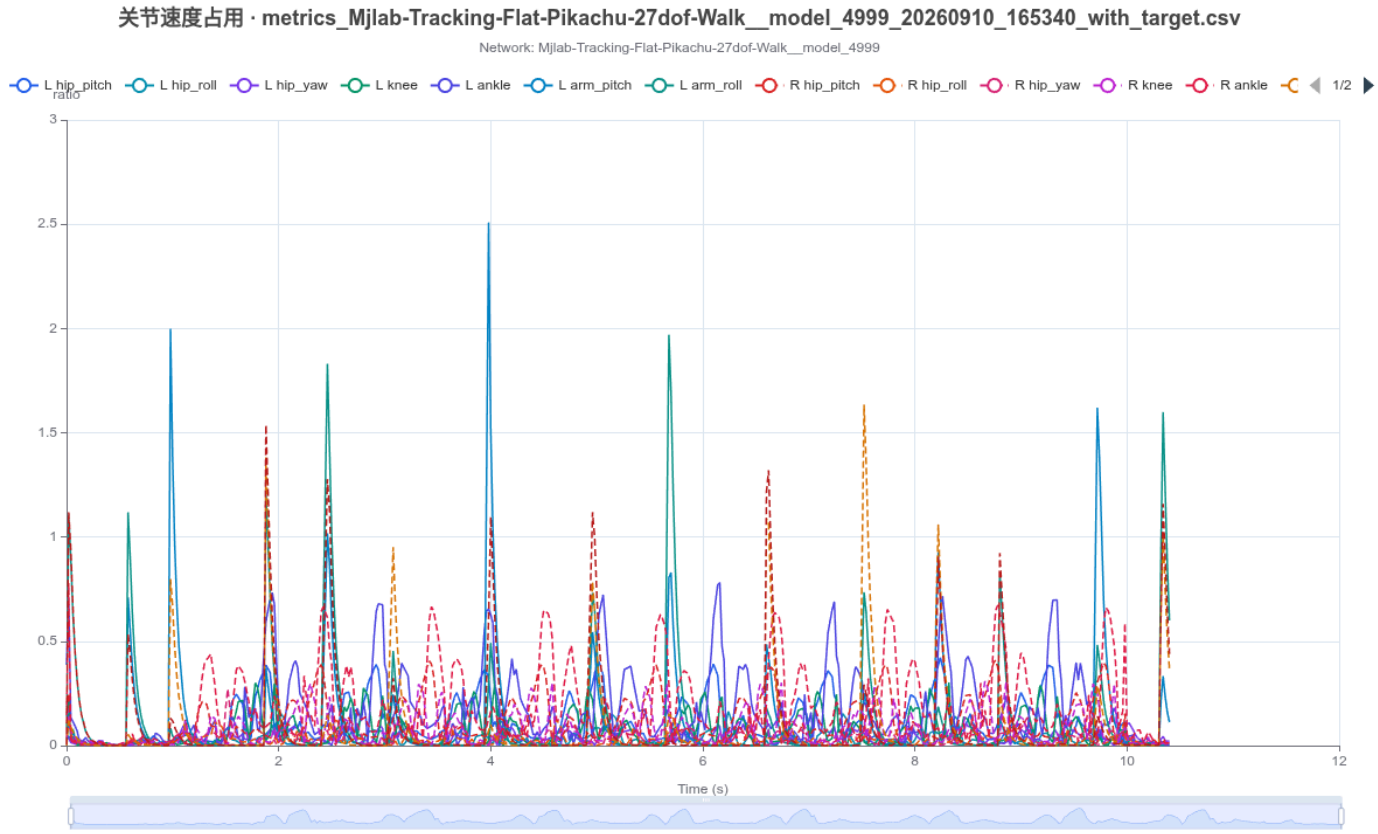


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_torque_ratio/left_hip_pitch_joint	30	10	454	0.0000849293	0.657379	0.138365	0.196187	0.657379	0.481603
joint_torque_ratio/left_hip_roll_joint	10	15	454	0.0000548749	0.134590	0.0345307	0.0487112	0.134590	0.107199
joint_torque_ratio/left_hip_yaw_joint	10	15	454	0.0000539243	0.128349	0.0223297	0.0309293	0.128349	0.0669585
joint_torque_ratio/left_knee_joint	30	10	454	0.000988446	0.272034	0.0796350	0.101947	0.272034	0.198074
joint_torque_ratio/left_ankle_joint	10	15	454	0.0000271294	0.242162	0.0708043	0.0881367	0.242162	0.181113
joint_torque_ratio/left_arm_pitch_joint	10	15	454	0.00000843731	1.00000	0.0126518	0.0603169	1.00000	0.0427690
joint_torque_ratio/left_arm_roll_joint	10	15	454	0.000170121	0.984345	0.0149222	0.0535958	0.984345	0.0516940
joint_torque_ratio/right_hip_pitch_joint	30	10	454	0.000152216	0.622078	0.127049	0.183746	0.622078	0.421217
joint_torque_ratio/right_hip_roll_joint	10	15	454	0.0000203695	0.140220	0.0338223	0.0519765	0.140220	0.124886
joint_torque_ratio/right_hip_yaw_joint	10	15	454	0.0000562822	0.0884179	0.0179603	0.0248387	0.0884179	0.0518764
joint_torque_ratio/right_knee_joint	30	10	454	0.000228883	0.327914	0.0670388	0.0837284	0.327914	0.166086
joint_torque_ratio/right_ankle_joint	10	15	454	0.0000221447	0.220062	0.0610534	0.0817248	0.220062	0.179845
joint_torque_ratio/right_arm_pitch_joint	10	15	454	0.00000751786	0.644030	0.0103609	0.0472089	0.644030	0.0427155
joint_torque_ratio/right_arm_roll_joint	10	15	454	0.0000958098	0.715544	0.0162450	0.0601942	0.715544	0.0533879

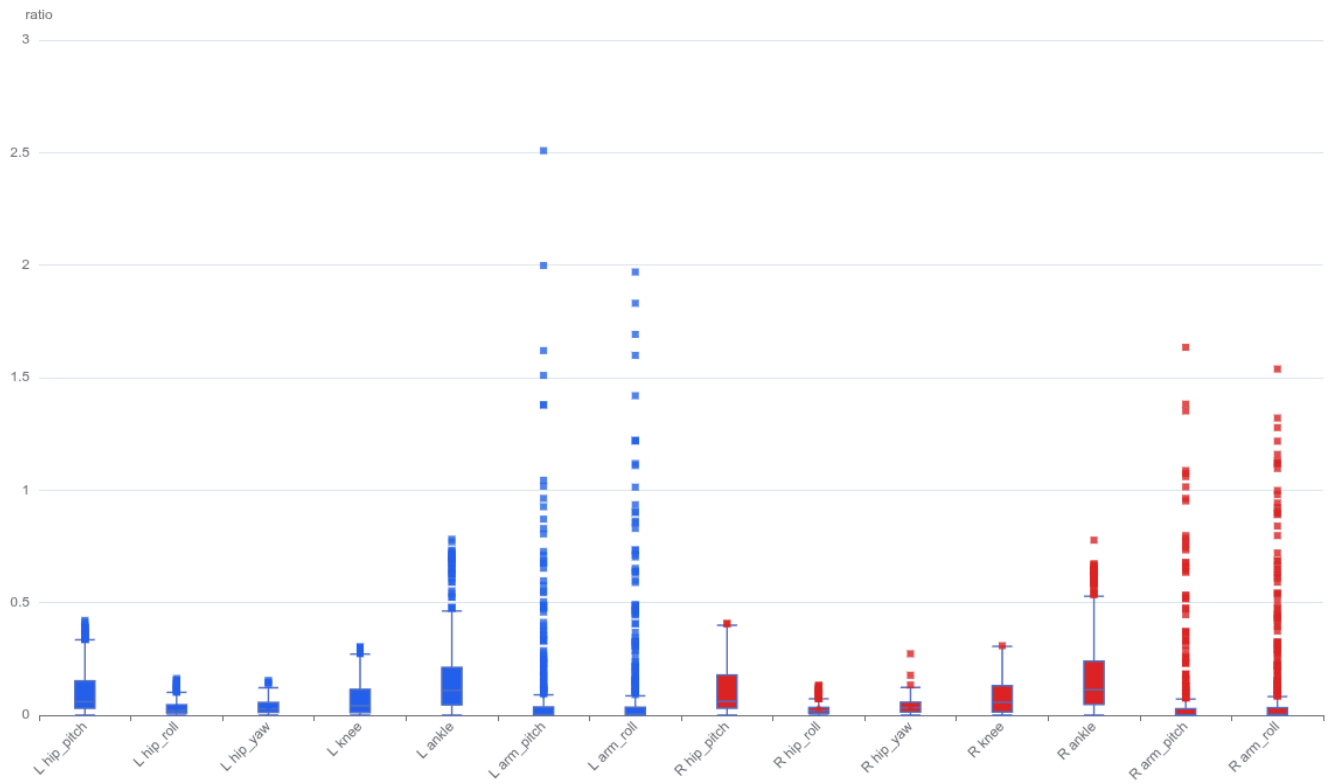

ratio
1
0.8
0.6
0.4
0.2
0
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

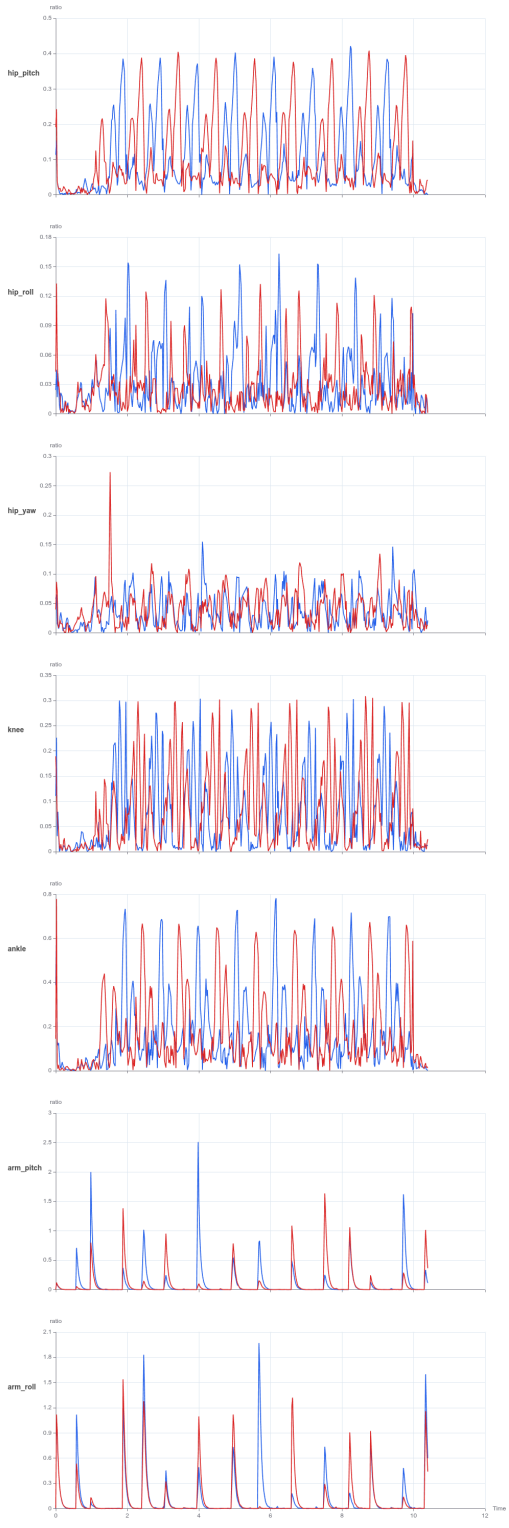

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
ratio
Time

关节速度占用

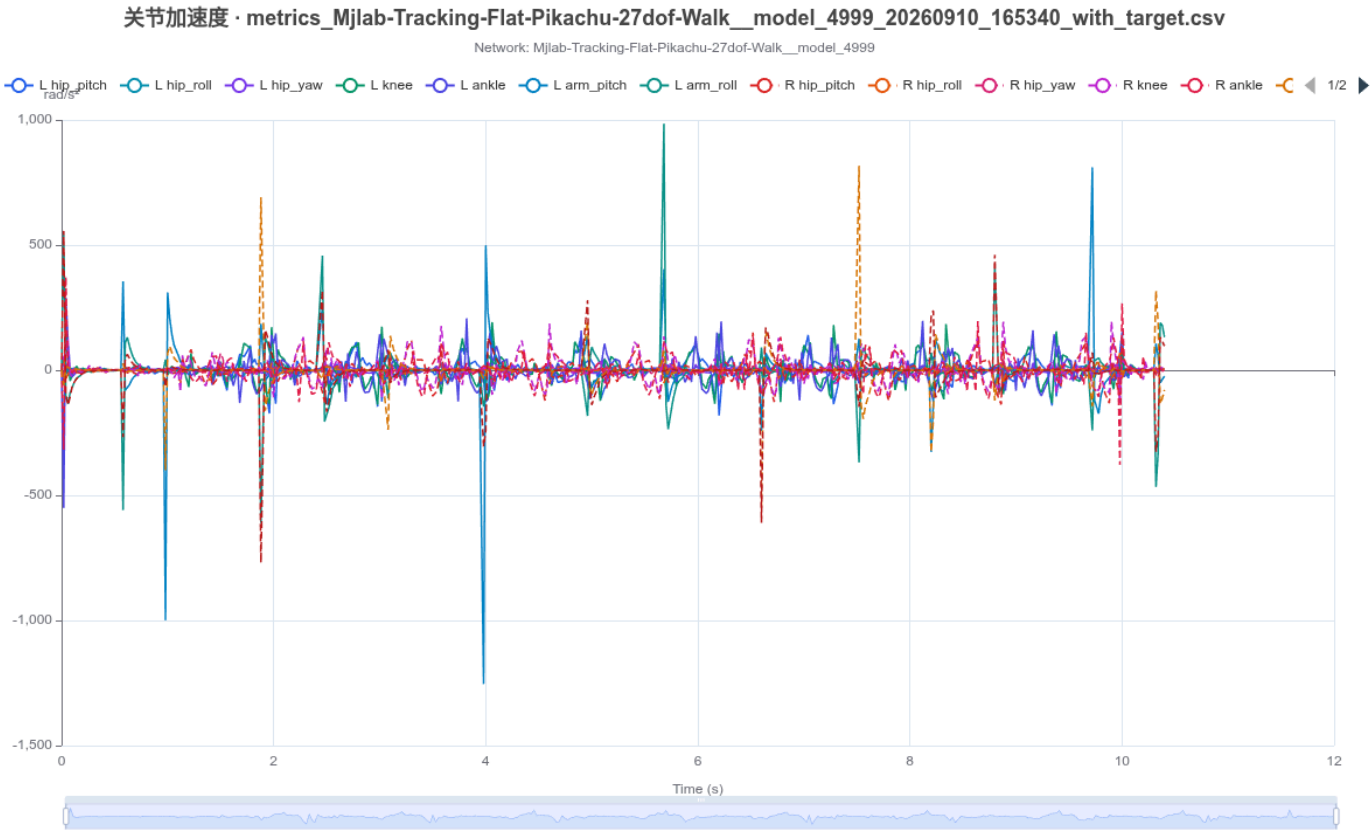


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_velocity_ratio/left_hip_pitch_joint	30	10	454	0.000124011	0.421415	0.105184	0.148221	0.421415	0.337529
joint_velocity_ratio/left_hip_roll_joint	10	15	454	0.0000206014	0.163370	0.0333308	0.0473779	0.163370	0.109009
joint_velocity_ratio/left_hip_yaw_joint	10	15	454	0.000144699	0.154800	0.0364040	0.0467377	0.154800	0.0927131
joint_velocity_ratio/left_knee_joint	30	10	454	0.00000520772	0.303472	0.0731178	0.105165	0.303472	0.229947
joint_velocity_ratio/left_ankle_joint	10	15	454	0.0000396824	0.781745	0.172555	0.247587	0.781745	0.628416
joint_velocity_ratio/left_arm_pitch_joint	10	15	454	6.51425e-7	2.50854	0.0901400	0.272074	2.50854	0.504187
joint_velocity_ratio/left_arm_roll_joint	10	15	454	0.00000254809	1.96940	0.100118	0.283477	1.96940	0.638797
joint_velocity_ratio/right_hip_pitch_joint	30	10	454	0.000486187	0.408564	0.107056	0.149271	0.408564	0.335696
joint_velocity_ratio/right_hip_roll_joint	10	15	454	0.000141367	0.132829	0.0278217	0.0392469	0.132829	0.0932267
joint_velocity_ratio/right_hip_yaw_joint	10	15	454	0.000114210	0.272942	0.0390409	0.0499250	0.272942	0.0963699
joint_velocity_ratio/right_knee_joint	30	10	454	0.0000566852	0.308649	0.0825061	0.115256	0.308649	0.252839
joint_velocity_ratio/right_ankle_joint	10	15	454	0.000181446	0.777816	0.177237	0.251284	0.777816	0.608343
joint_velocity_ratio/right_arm_pitch_joint	10	15	454	0.0000224055	1.63471	0.0753431	0.223590	1.63471	0.520247
joint_velocity_ratio/right_arm_roll_joint	10	15	454	0.00000841051	1.53820	0.0940667	0.255819	1.53820	0.633407

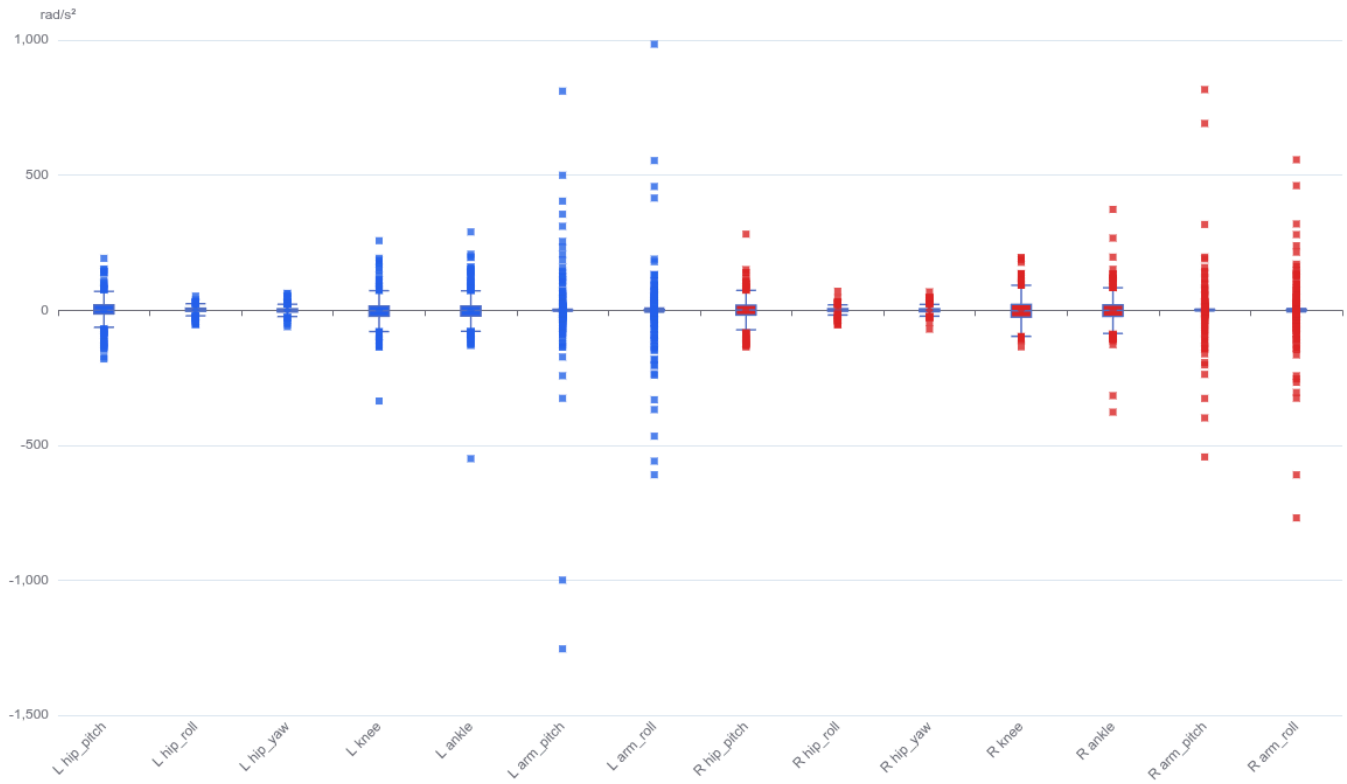

ratio
3
2.5
2
1.5
1
0.5
0
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

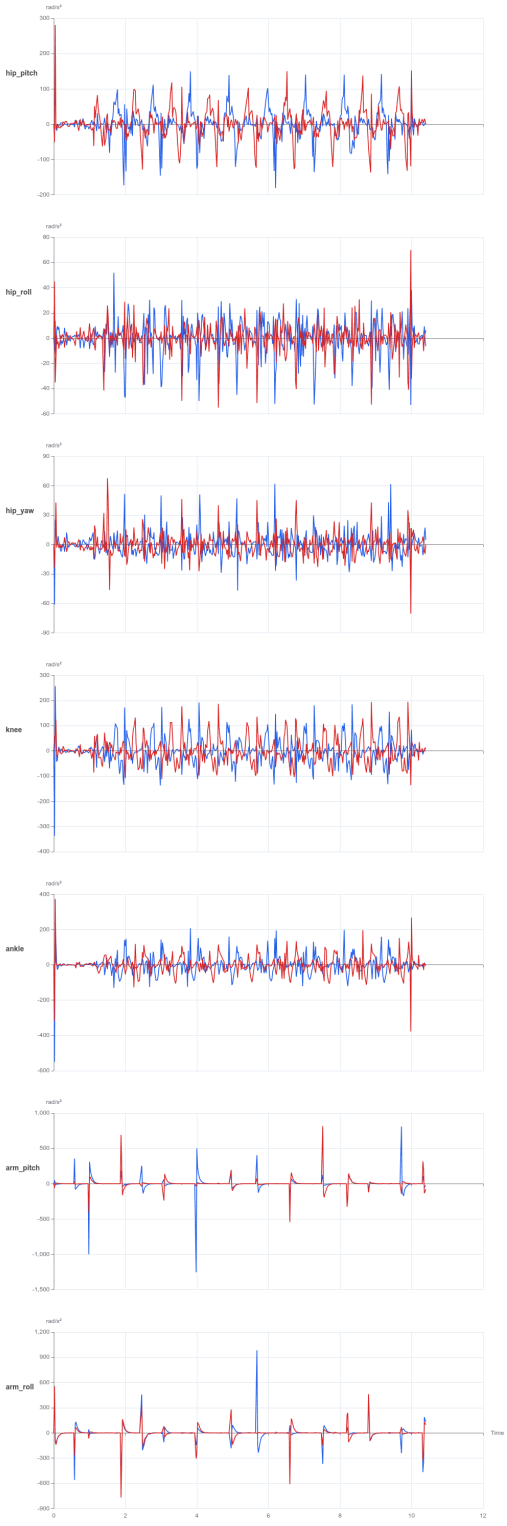

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
ratio
Time

关节加速度

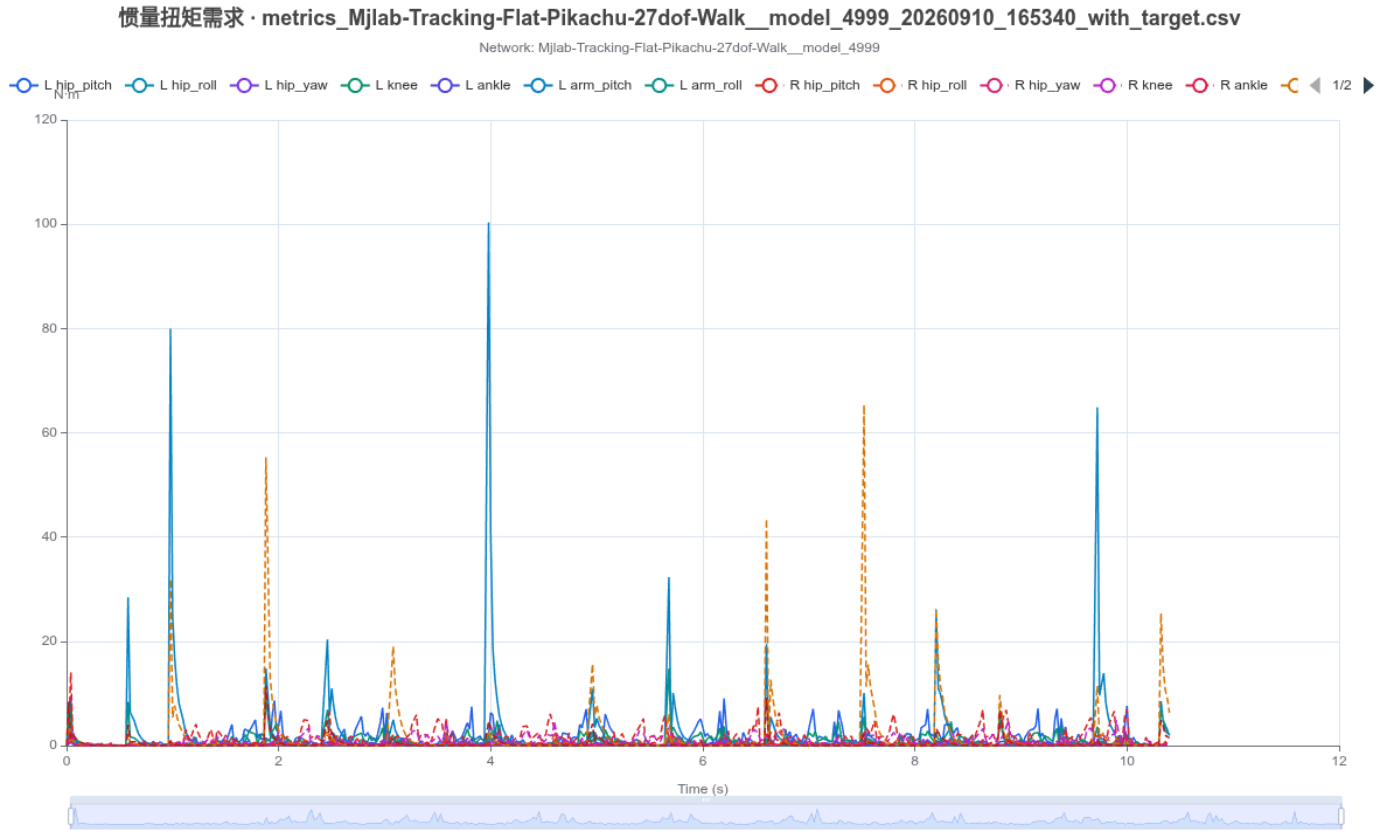


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_accel_rad_s2/left_hip_pitch_joint	30	10	454	-180.888	191.669	0.134379	42.6914	191.669	98.5287
joint_accel_rad_s2/left_hip_roll_joint	10	15	454	-52.9898	51.9338	0.172086	14.0129	52.9898	29.8586
joint_accel_rad_s2/left_hip_yaw_joint	10	15	454	-61.3058	61.8458	-0.0333414	12.4776	61.8458	23.6590
joint_accel_rad_s2/left_knee_joint	30	10	454	-336.587	256.961	-0.0335158	53.2334	336.587	110.397
joint_accel_rad_s2/left_ankle_joint	10	15	454	-550.076	289.685	-0.669440	55.9650	550.076	106.533
joint_accel_rad_s2/left_arm_pitch_joint	10	15	454	-1254.31	810.844	-0.385778	100.803	1254.31	111.653
joint_accel_rad_s2/left_arm_roll_joint	10	15	454	-609.551	984.768	-1.22834	88.1215	984.768	131.161
joint_accel_rad_s2/right_hip_pitch_joint	30	10	454	-136.598	281.401	0.158269	43.0678	281.401	97.2006
joint_accel_rad_s2/right_hip_roll_joint	10	15	454	-55.1036	69.8943	0.0367948	11.8815	69.8943	25.5523
joint_accel_rad_s2/right_hip_yaw_joint	10	15	454	-70.4594	67.6072	0.0933511	11.8872	70.4594	22.7687
joint_accel_rad_s2/right_knee_joint	30	10	454	-136.045	194.216	0.877219	50.5821	194.216	108.281
joint_accel_rad_s2/right_ankle_joint	10	15	454	-377.671	372.857	0.298269	53.1540	377.671	98.8633
joint_accel_rad_s2/right_arm_pitch_joint	10	15	454	-543.736	816.660	0.914473	72.5642	816.660	108.584
joint_accel_rad_s2/right_arm_roll_joint	10	15	454	-769.341	556.842	-1.71108	76.4732	769.341	128.022

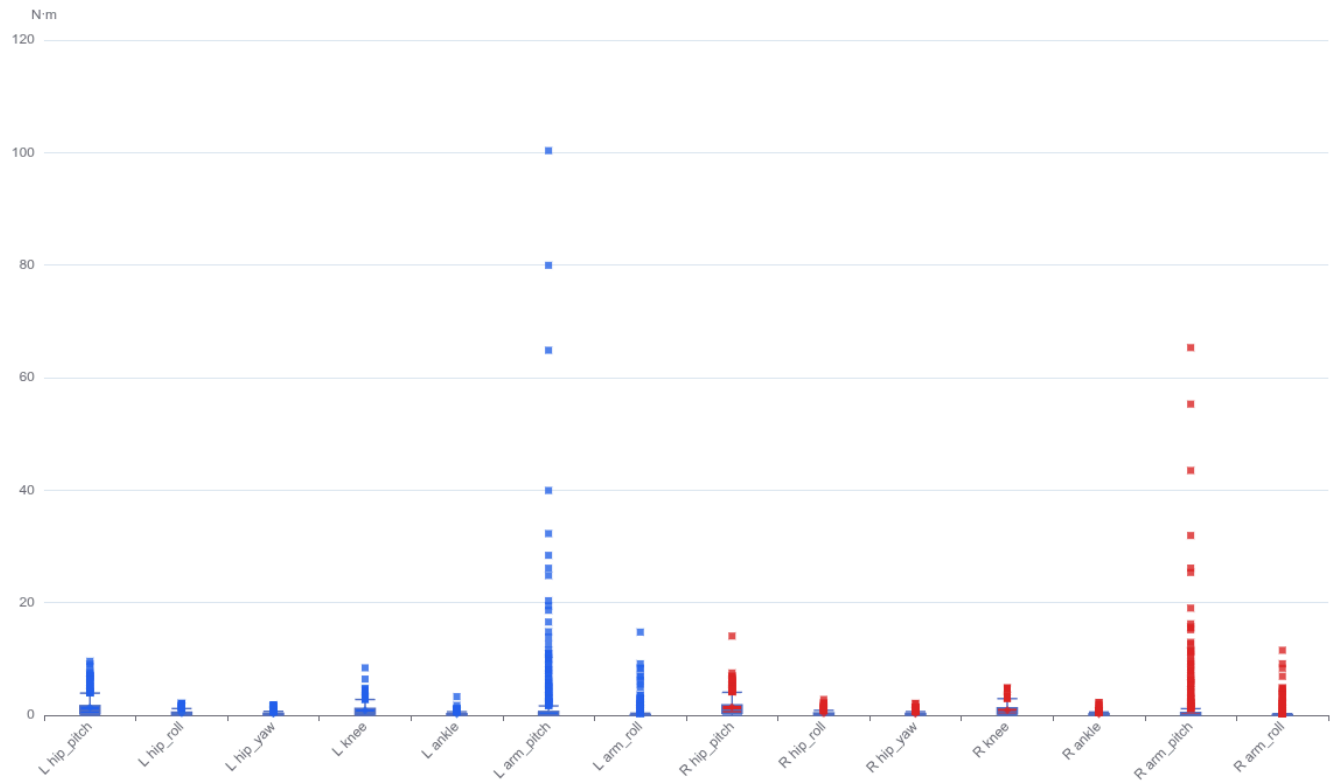

rad/s²
1,000
500
0
-500
-1,000
-1,500
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

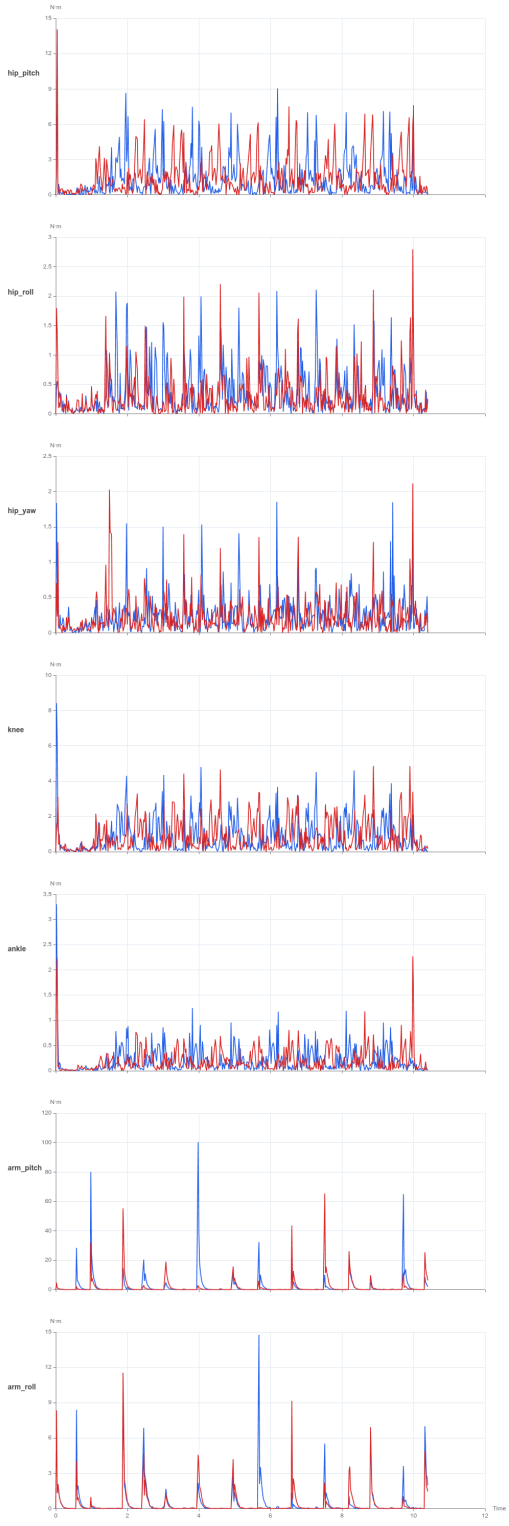

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
Time

惯量扭矩需求

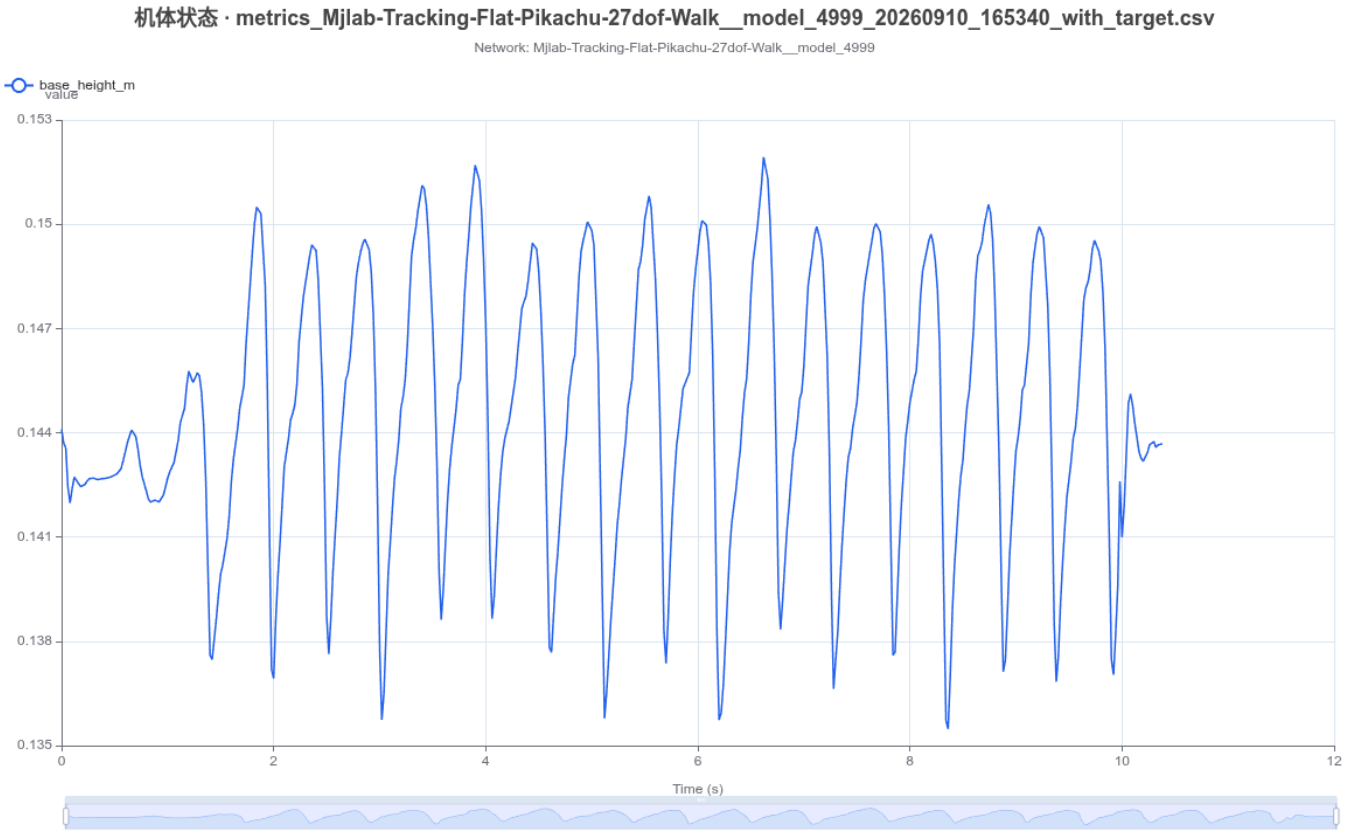


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_inertia_torque_nm/left_hip_pitch_joint	30	10	454	0.00000	9.58347	1.35950	2.13457	9.58347	4.92643
joint_inertia_torque_nm/left_hip_roll_joint	10	15	454	0.00000	2.11959	0.378615	0.560518	2.11959	1.19434
joint_inertia_torque_nm/left_hip_yaw_joint	10	15	454	0.00000	1.85537	0.249693	0.374329	1.85537	0.709769
joint_inertia_torque_nm/left_knee_joint	30	10	454	0.00000	8.41469	0.868032	1.33083	8.41469	2.75993
joint_inertia_torque_nm/left_ankle_joint	10	15	454	0.00000	3.30046	0.198221	0.335790	3.30046	0.639196
joint_inertia_torque_nm/left_arm_pitch_joint	10	15	454	0.00000	100.345	1.97773	8.06426	100.345	8.93223
joint_inertia_torque_nm/left_arm_roll_joint	10	15	454	0.00000	14.7715	0.389483	1.32182	14.7715	1.96742
joint_inertia_torque_nm/right_hip_pitch_joint	30	10	454	0.00000	14.0700	1.43744	2.15339	14.0700	4.86003
joint_inertia_torque_nm/right_hip_roll_joint	10	15	454	0.00000	2.79577	0.300123	0.475261	2.79577	1.02209
joint_inertia_torque_nm/right_hip_yaw_joint	10	15	454	0.00000	2.11378	0.237864	0.356616	2.11378	0.683062
joint_inertia_torque_nm/right_knee_joint	30	10	454	0.00000	4.85541	0.893014	1.26455	4.85541	2.70704
joint_inertia_torque_nm/right_ankle_joint	10	15	454	0.00000	2.26602	0.198707	0.318924	2.26602	0.593180
joint_inertia_torque_nm/right_arm_pitch_joint	10	15	454	0.00000	65.3328	1.55775	5.80514	65.3328	8.68669
joint_inertia_torque_nm/right_arm_roll_joint	10	15	454	0.00000	11.5401	0.354958	1.14710	11.5401	1.92033

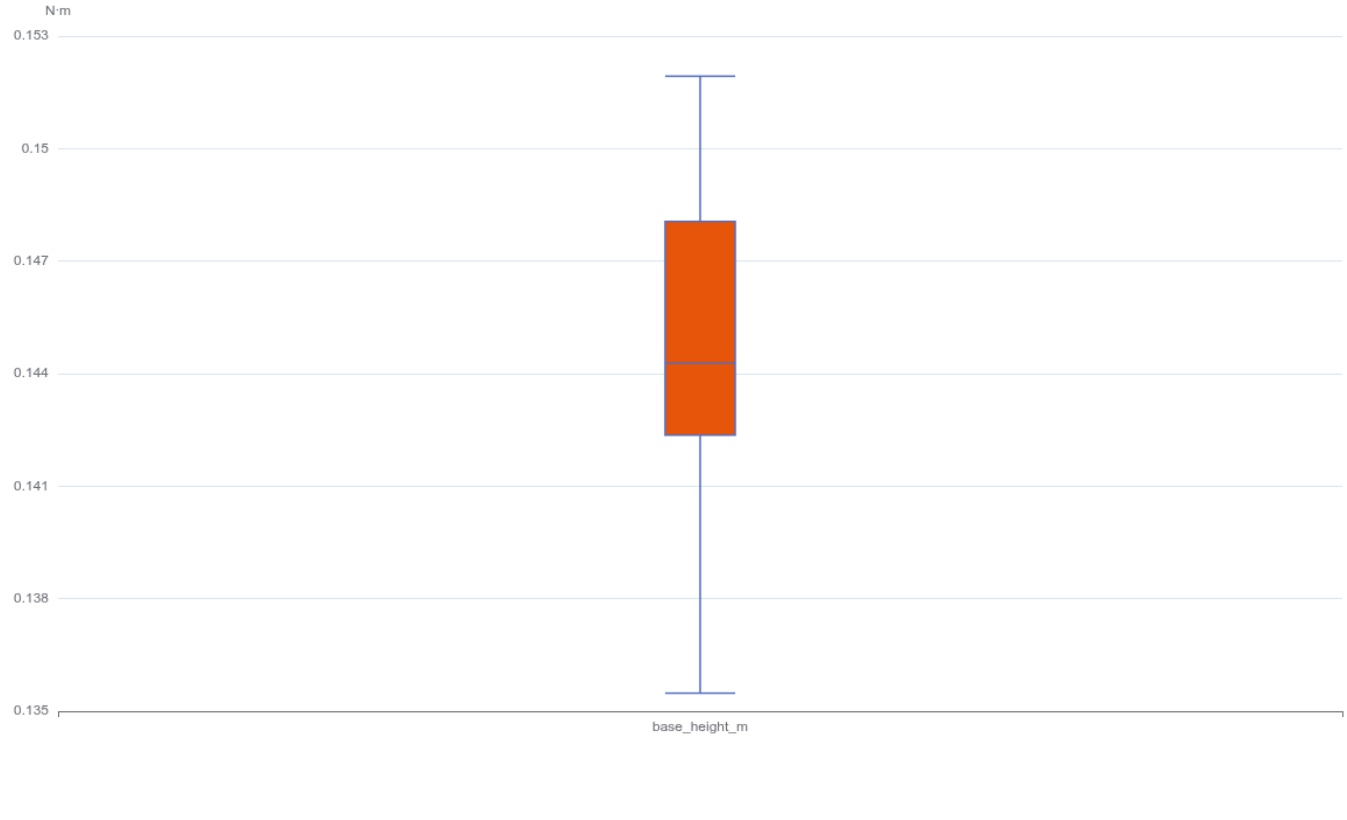

N·m
120
100
80
60
40
20
0
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll


hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
N·m
Time

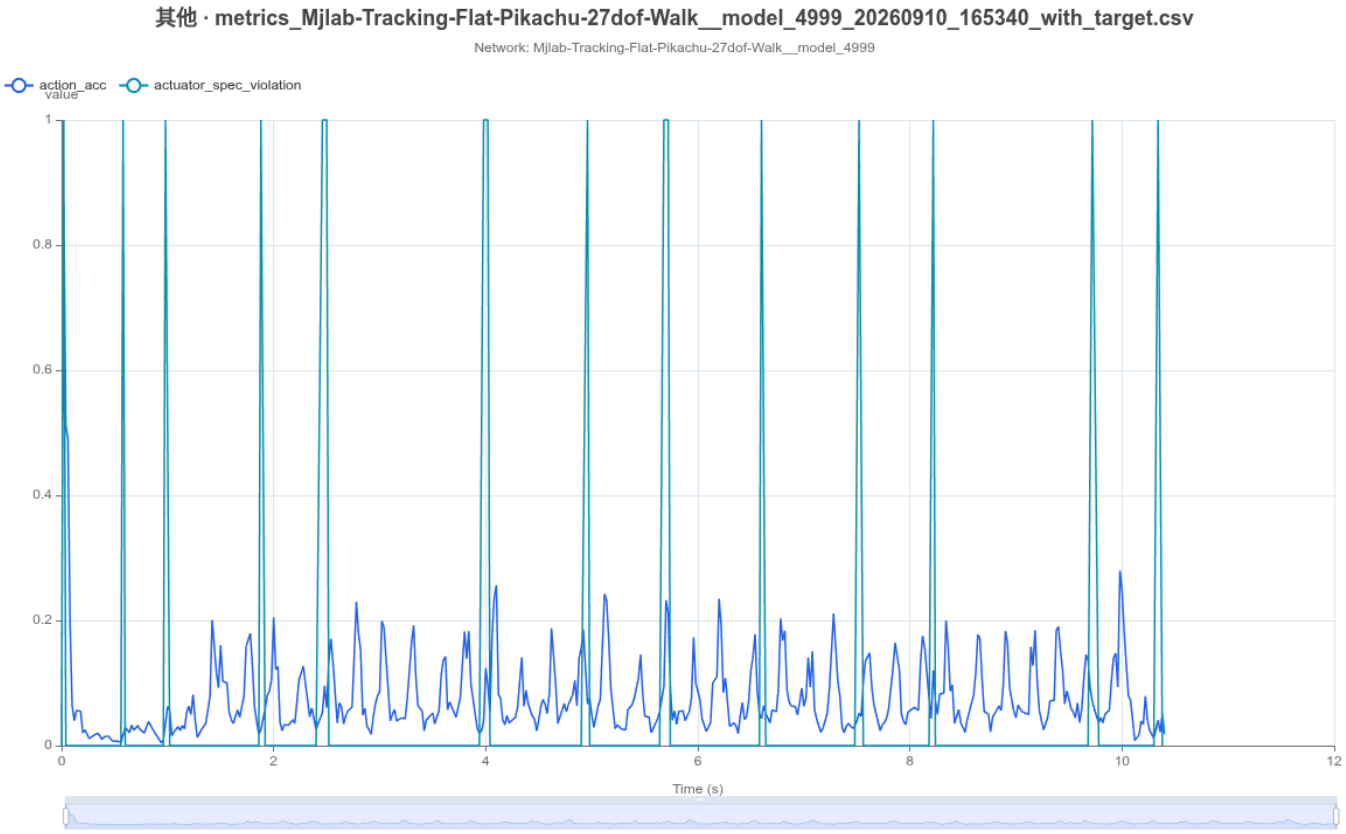
机体状态



Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
base_height_m	–	–	454	0.135481	0.151929	0.144559	0.144611	0.151929	0.150103



其他



Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
action_acc	–	–	454	0.00452189	0.880984	0.0768722	0.104617	0.880984	0.188569
actuator_spec_violation	–	–	454	0.00000	1.00000	0.0528634	0.229920	1.00000	1.00000

